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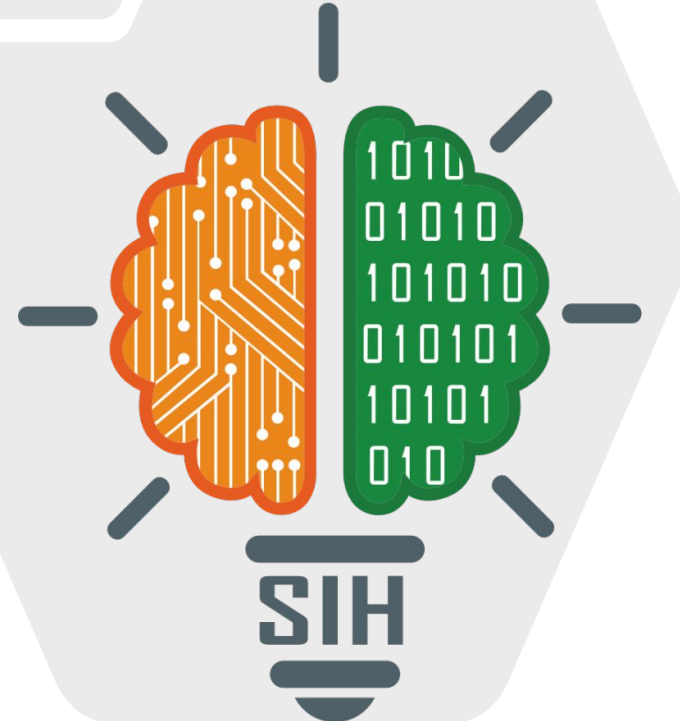
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TITLE PAGE

SMART INDIA HACKATHON 2026

- Problem Statement ID - 26095
- Problem Statement Title- Smart Real-Time
Monitoring & Inspection Mobile App
- Title- DRISHTI
- Name- Team Cortex





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DRISHTI

PROPOSED SOLUTION



MoSJE Oversight & Web Dashboard
Define Policy, Monitor



Central AI Hub & Automation
Data Process, Random Assign



Analytics & Insights Detect
Anomalies, Verify Attendance



Field App & Live Feeds
Capture Geo- Evidence, Live VC

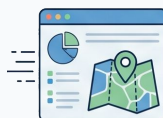
HOW IT ADDRESSES THE PROBLEM



Eliminates Fake Reporting
Tamper-proof Data



Prevents Collusion
Random Assignments

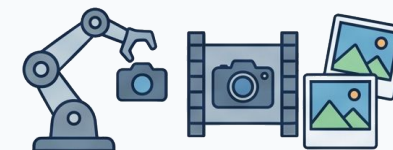


Overcomes Bottlenecks
Efficient Scalability



Automates Compliance
24/7 AI Auditing

INNOVATION & UNIQUENESS



Lightweight AI
Efficient, Low-Bandwidth Vision



Zero-Trust Evidence
Hardware-Verified Data



Unified Ecosystem Centralized
Monitoring Platform



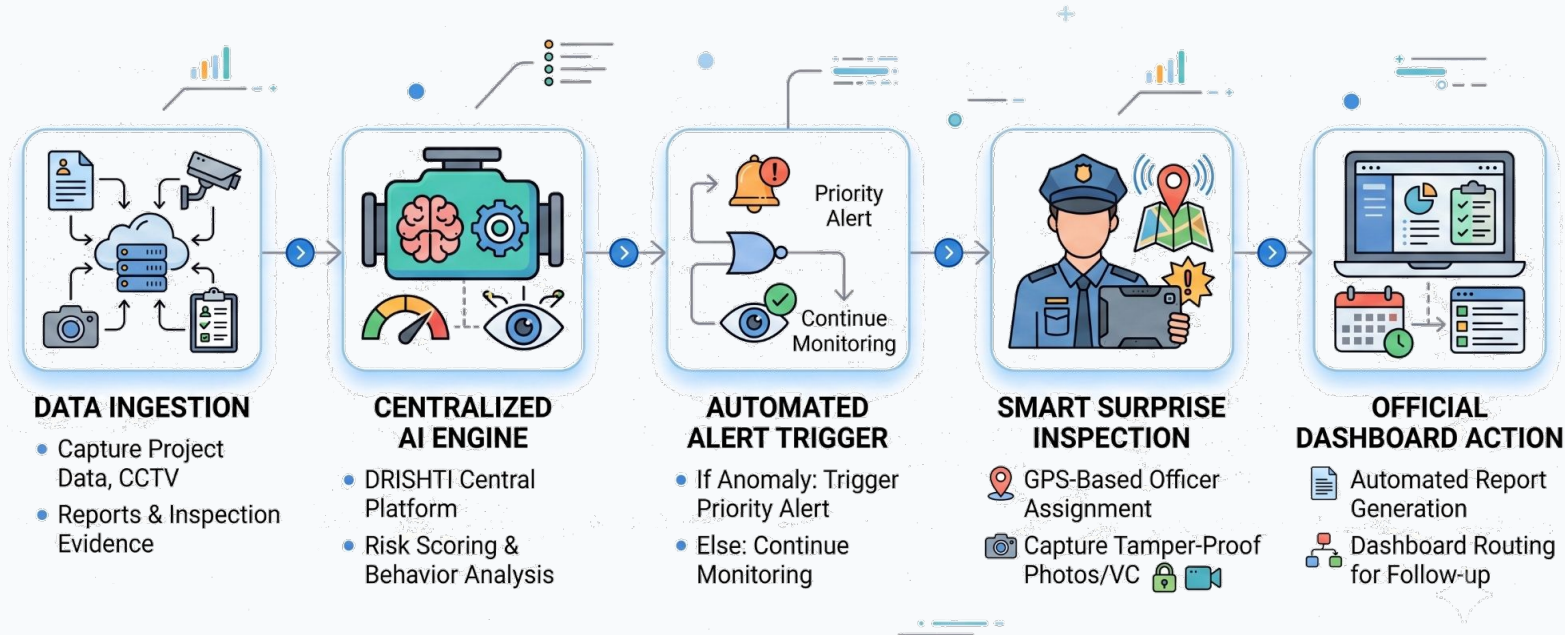
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TECHNICAL APPROACH

METHODOLOGY & FLOW



TECH STACK

	Field App Flutter (Dart)	
	Dashboard React + Vite	
	Backend FastAPI + PostgreSQL	
	AI Layer IsolationForest + OpenCV	
	Maps Leaflet + OpenStreetMap	
	Video / VC HLS feed + Jitsi Meet	
	Auth & Geo JWT (3 roles) + GPS	

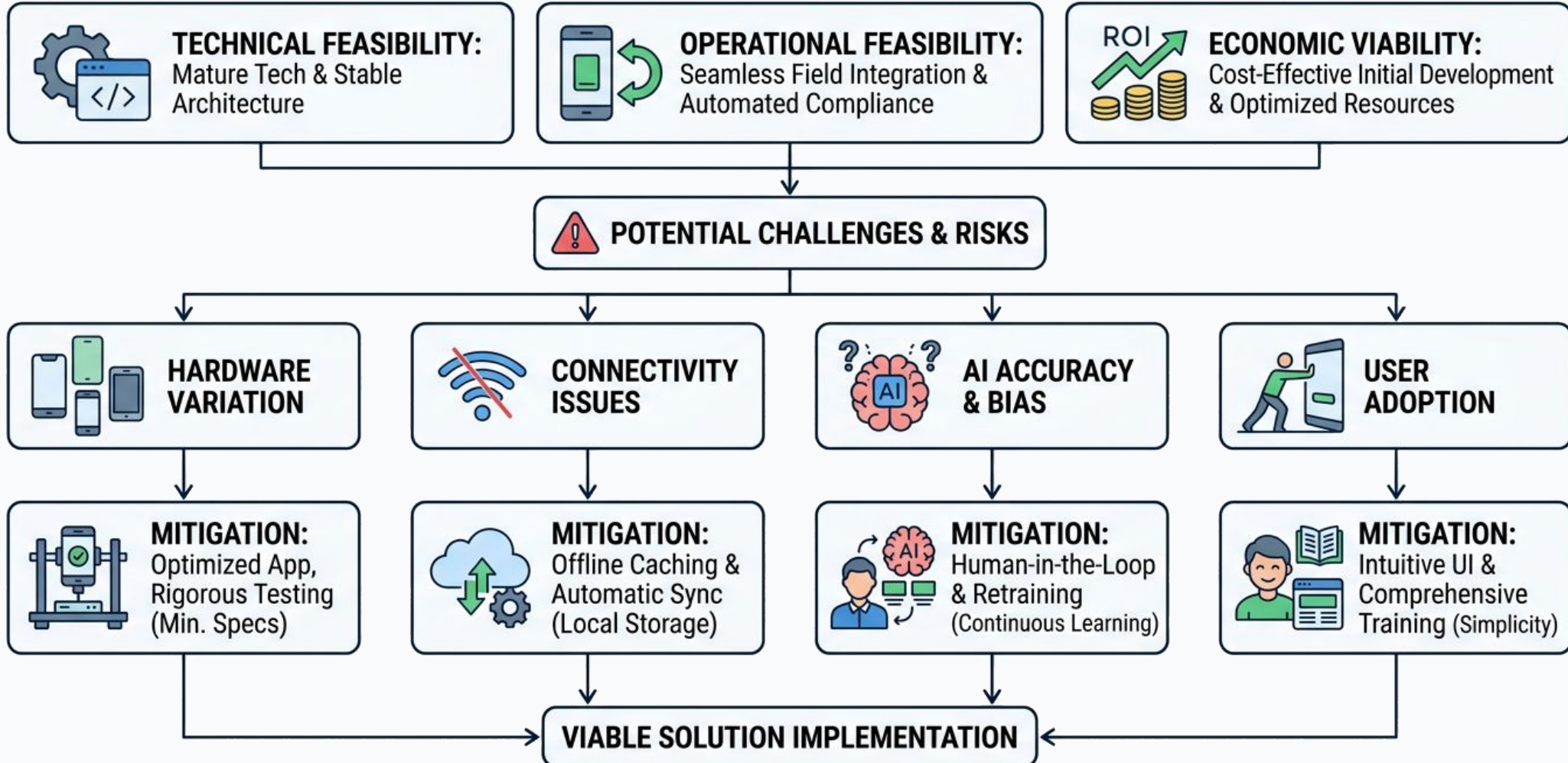


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FEASIBILITY & VIABILITY





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MULTIDISCIPLINARY REQUIREMENTS

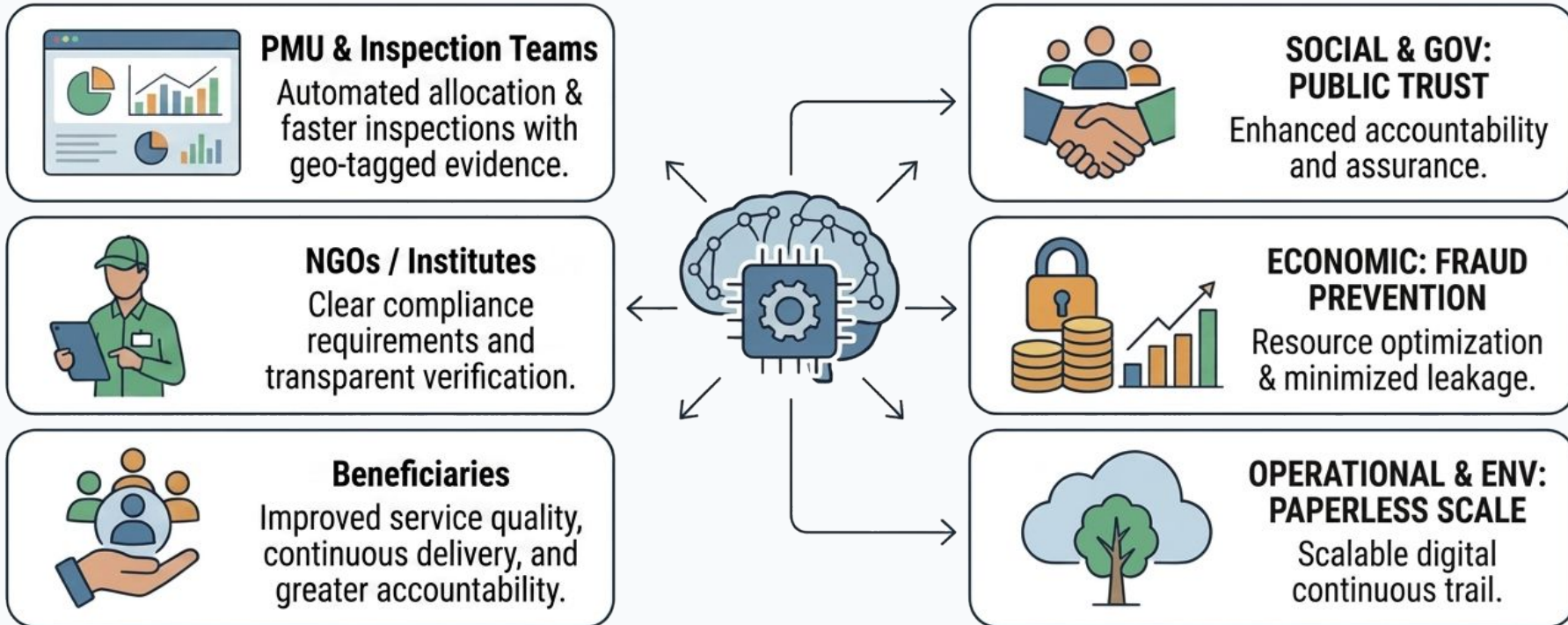
Part of the Problem Statement requiring other core branch support	Branch	Problem solution
Live CCTV feed integration & camera/IoT hardware compatibility across institute sites	Electronics & Communication Engineering (ECE)	ECE support to standardize camera/IoT interfacing (ONVIF/RTSP protocol compliance), handle video signal processing, and manage hardware-software integration at edge devices
Power backup for CCTV/IoT devices at institute & NGO sites	Electrical & Electronics Engineering (EEE)	Design backup power solutions (battery/solar) to keep cameras and sensors operational during outages
Real-time video streaming & random VC connectivity in low-network/rural areas	Electronics & Telecommunication Engineering (ETE)	Optimize bandwidth usage, adaptive bitrate streaming, and network protocol/handoff (4G/5G/Wi-Fi) design for reliable low-latency video in poor-connectivity zones



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IMPACT & BENEFITS



FROM:  REPORT-BASED INSPECTION → TO:  EVIDENCE-BASED CONTINUOUS VERIFICATION



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RESEARCH OF SIMILAR SOLUTIONS

Existing Solution	Advantages	Disadvantages
PRAGATI (PMO/NIC, national project-monitoring platform)	Enables real-time monitoring of major government projects through PM-chaired reviews with video conferencing, geo-spatial mapping, and drone feeds; has monitored 3,300+ projects worth over ₹85 lakh crore with a 90%+ issue-resolution rate	Built for large infrastructure/ministry-level projects reviewed monthly at the top, not designed for continuous, ground-level surveillance of individual institutes, NGOs, or field staff; no live CCTV, no random inspection assignment, no AI anomaly detection
PM POSHAN Automated Monitoring System + state inspection apps (e.g. WB's MDM-Inspection app)	Uses IVRS/SMS/mobile apps to collect real-time data from schools with no internet dependency, and dedicated inspection apps exist specifically to identify irregularities at the school level	Relies heavily on self-reported data from the same institution being monitored, no independent video verification; inspections aren't AI-randomized, so visits remain predictable and can be anticipated by staff
SafetyCulture / iAuditor (now Mitti), global commercial inspection app	Lets teams digitize inspection forms, schedule recurring inspections, assign follow-up tasks, and log real-time issues, with photo and video evidence capture even offline; trusted by 70,000+ organizations	General-purpose commercial tool, not built for government welfare-scheme oversight; GPS tracking is still in early access and unreliable; subscription-based licensing, no live CCTV feed or surprise-VC capability, inspections scheduled by managers rather than randomized



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REFERENCES & RESEARCH — DRISHTI



Official Government References

Ministry of Social Justice & Empowerment

Official social welfare initiatives & policies

<https://socialjustice.gov.in/>

Dept of Social Justice & Empowerment

Program execution & welfare schemes monitoring

<https://socialjustice.gov.in/schemes>

Digital India Portal

Digital governance & service delivery standards

<https://digitalindia.gov.in/>

UIDAI Aadhaar & e-KYC

Secure beneficiary identity verification

<https://uidai.gov.in/>



Technical Documentation References

OpenCV Framework

Computer vision processing algorithms

<https://docs.opencv.org/>

Ultralytics YOLO Engine

Real-time object detection & site monitoring

<https://docs.ultralytics.com/>

WebRTC Protocol

Real-time low-latency video calls & checks

<https://webrtc.org/>

PostgreSQL DB Engine

Scalable relational database infrastructure

<https://postgresql.org/docs>